



# Flow Characteristics of a Recirculating Flameless Combustor Configuration

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The flameless combustion process has been shown in industrial burners and power generators to substantially reduce pollutant emissions, however, its application to propulsive gas turbines has yet to be proven. Experimental research at the University of Cincinnati has demonstrated flameless combustion in a device aimed at being used in a propulsive engine. The combustion chamber relies on fluid dynamics to create the mixing necessary to achieve flameless combustion. In this paper, we present an investigation of this recirculating flameless combustor. We use the Naval Research Laboratory's Discontinuous-Galerkin code, *Propel*, to simulate the recirculation of a reacting gaseous mixture with high accuracy in space and time. The goal of this work is to use diagnostics from these simulations to better understand the mechanisms and conditions that must be present for a sustained flameless combustion process.

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Table 1. List of Transport, Thermodynamic, and Chemistry definitions

| Term           | Comments                                                   |
|----------------|------------------------------------------------------------|
| $\mathbf{v}$   | Velocity                                                   |
| $\rho$         | Density                                                    |
| $T$            | Temperature                                                |
| $p$            | Pressure                                                   |
| $\mathbb{T}$   | Deviatoric stress tensor                                   |
| $\mu$          | Viscosity coefficient                                      |
| $\lambda$      | Thermal Conductivity                                       |
| $u$            | internal energy                                            |
| $C_k$          | Concentration of species $k$                               |
| $\mathbf{V}_k$ | Diffusion velocity of species $k$                          |
| $\omega_k$     | Concentration production rate of species $k$               |
| $D_k$          | Diffusion coefficient of species $k$                       |
| $h_k$          | Enthalpy of species $k$                                    |
| $m_k$          | Molecular weight of species $k$                            |
| $e_t$          | Total energy, $u + \frac{1}{2}\mathbf{v} \cdot \mathbf{v}$ |
| $\alpha$       | Thermal diffusivity, $\lambda / (C_p \rho)$                |

## I. Introduction

Igniting fuels with preheated air, typically in excess of 1000°C, in combination with a recirculation process that introduces combustion products near fuel injection results in a stabilized combustion processes. This combination of high temperature air and recirculation results in a distributed homogeneous burning which does not produce a visible flame front and has thus been referred to as flameless combustion. Other researchers have called it Excess Enthalpy Combustion (EEC), High Temperature Air Combustion (HiTAC), Moderate and Intense Low Oxygen Diffusion (MILD), Flameless Oxidation (FLOX), and Colorless Distributed Combustion (CDC). Due to the absence of a flame, flameless combustion has been experimentally demonstrated to lead to less production of pollutants such as nitric oxide and unburned hydrocarbons.<sup>1–5</sup>

The potential of significantly reducing pollutants highlight the promise of flameless combustion and the technology has already been applied to stationary combustion facilities. Although this technology is used on ground energy generators, it has not migrated into propulsion system designs. This is partly due to the complexity of a recirculating combustor and unknown system performances and weight penalties. It is possible that flameless combustion throughout the operating range of a typical gas-turbine propulsive engine is not feasible. However, the beneficial characteristics of flameless combustion could be used to varying degrees if the physics and chemistry behind this concept were better understood.

Recently, the University of Cincinnati<sup>6–11</sup> has shown experimentally that recirculation can be achieved by using fluid dynamic control by introducing swirl in the preheated air and angled fuel injectors. This paper focuses on examining the characteristics of flow recirculation in the University of Cincinnati's flameless combustor experimental configuration. Particularly we will focus on the importance of fuel and hot air injector geometry as well as examine the importance of heat release on flameless combustion with simple chemistry models. In this manuscript we present the time-averaged and time-accurate results of a high order, finite element method code, Propel, that was used to simulate the entire experimental configuration. Our ongoing work is investigating various operating conditions and relevant geometric configurations to understand the fluid dynamic mechanisms that resulted in the effective circulation for flameless combustion.

## II. Experimental Configuration

Figure 1 shows the experimental configuration that resulted in effective recirculation that led to flameless combustion. The combustor is aligned vertically with a hot air bath, 700-1100 °C, at pressures ranging from 1.2-1.3 atm, depending on the flow-rate conditions. Hot air, depicted as blue in the diagram, travels through

side veins that circulate in an inner bowl while rising to the main level. The air circulation within the bowl is swirled due to the angled veins as depicted in the bottom right of the figure. The hot swirling air then travels to an upper level where fuel is injected, depicted as red in the diagram, by curved tube injectors. The fuel then penetrates the air creating a combustion region, which is depicted as orange. Some combustion products and intermediate species are recirculated towards the injection region, depicted as dashed orange lines, encouraging the sustained flameless combustion whereas other products leave the system as exhaust, shown as green lines. This experimental configuration was able to produce flameless combustion and flames depending on the equivalence ratio, while having minimal pressure loss through the injector system. The details of the fluid dynamics within the combustion chamber, however, are not fully understood, and the main goal of this ongoing research is to better explain this recirculation pattern, especially as the geometry and conditions are changed.

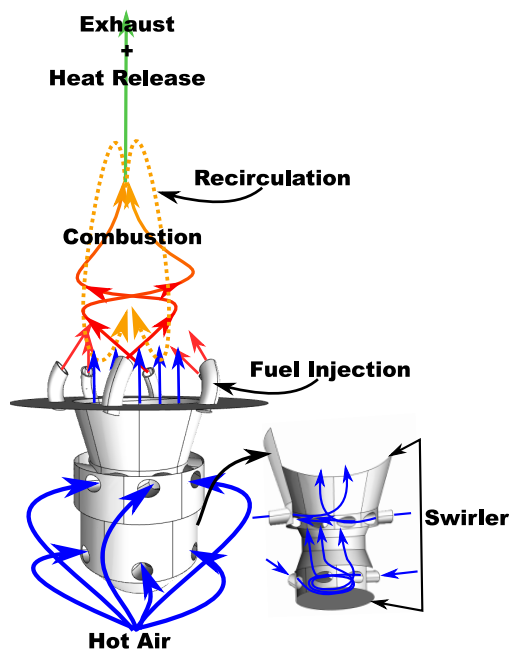


Figure 1. Experimental combustor with labeled regions.

It was found in the University of Cincinnati experiments that achieving flameless combustion is highly dependent on the angle and shape of the fuel injectors. Already depicted are injectors that are angled towards the swirling hot air flow at a 20-degree offset from the vertical. The University of Cincinnati experimentalists concluded that fuel penetration was key to recirculation and this manuscript focuses on the fuel penetration and mechanisms that convected the fuel into the recirculation zone.

### III. Numerical Approach and Methodology

#### A. Conservation Equations

The Naval Research Laboratory has an in-house combustion code, Propel, that uses the discontinuous-Galerkin (DG) finite element method (FEM) to solve the following conservation equations:

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v}) = -\nabla p - \nabla \cdot \mathbb{T}(\mu, \nabla \mathbf{v}) \quad (1)$$

$$\begin{aligned} \frac{\partial \rho e_t}{\partial t} + \nabla \cdot ((\rho e_t + p) \mathbf{v}) &= \nabla \cdot \lambda \nabla T \\ &\quad - \nabla \cdot \sum_{i=1}^{N_s} \rho h_k D_k \nabla C_k \\ &\quad - \nabla \cdot (\mathbb{T}(\mu, \nabla \mathbf{v}) : \mathbf{v}) \end{aligned} \quad (2)$$

$$\frac{\partial C_k}{\partial t} + \nabla \cdot (C_k \mathbf{v}) = \nabla \cdot D_k \nabla C_k + \omega_k. \quad (3)$$

Density, pressure, and internal energy are calculated based on concentrations and mixture temperature by  $\rho = \sum_{k=1}^{N_s} C_k W_k$ ,  $p = R^o T \sum_{k=1}^{N_s} C_k$ , and  $e_t = u(C_k, T) + \frac{\mathbf{v} \cdot \mathbf{v}}{2}$  respectively. All transport quantities,  $D_k$ ,  $\lambda$ , and  $\mu$ , are evaluated as concentration based mixture-averaged quantities of polynomial fits based on temperature as provided by References 12–14. However, the polynomials were adjusted so that one 6<sup>th</sup> order polynomial, instead of two, can describe the thermodynamic state of each species within 0.01 percent accuracy of the SANDIA fits in temperature ranges 300 to 3000 K ensuring continuous differentiability in  $C^\infty$ . This ensures differentiability for implicit methods that we are currently developing. The reaction term,  $\omega_k$ , is calculated using a time-split approach that supports implicit detailed chemical mechanism integration in time.

## B. The discontinuous Galerkin discretization

Here we rewrite 1–3 in a more concise, flux-expression form, representation as

$$\frac{\partial \mathbf{y}}{\partial t} + \nabla \cdot (\mathcal{F}^c(\mathbf{y}) - \mathcal{F}^v(\mathbf{y})) = \mathbf{S}(\mathbf{y}). \quad (4)$$

Where

$$\mathbf{y} = \begin{Bmatrix} \rho \mathbf{v} \\ \rho \left(u + \frac{\mathbf{v} \cdot \mathbf{v}}{2}\right) \\ C_k \end{Bmatrix} \quad (5)$$

$$\mathcal{F}^c = \begin{Bmatrix} \rho \mathbf{v} \otimes \mathbf{v} - p \mathbb{I} \\ (\rho e_t + p) \mathbf{v} \\ C_k \mathbf{v} \end{Bmatrix}, \quad (6)$$

$$\mathcal{F}^v = \begin{Bmatrix} \mathbb{T}\left(\frac{\mu(C_i, T)}{Re_r}, \nabla \mathbf{v}\right) \\ \frac{1}{Pr_r Re_r} \lambda(C_i, T) \nabla T - \frac{1}{Re_r} (\mathbb{T}(\mu(C_i, T), \mathbf{v}) : \mathbf{v}) - \nabla \cdot \frac{1}{Re_r} (\mathbb{T}(\mu(C_i, T), \mathbf{v}) : \mathbf{v})) \\ \frac{1}{Re_r Sc_r} D_i \nabla C_k \end{Bmatrix}, \quad (7)$$

and

$$\mathbf{S}(\mathbf{y}) = \begin{Bmatrix} \mathbf{0} \\ 0 \\ \omega_k \end{Bmatrix} \quad (8)$$

The domain,  $\Omega$ , is decomposed into non-overlapping finite elements and denoted as  $\mathcal{T}_h$ . The discrete space of discontinuous polynomials is then defined as, for  $p \in \mathbb{N}$ ,

$$\mathbf{V}_h^p := \left\{ \mathbf{v}_h \in [L^2(\Omega)]^d : \mathbf{v}_{h|K} \in [\mathbb{P}_p(K)]^d \forall K \in \mathcal{T}_h \right\}, \quad (9)$$

We define the jump and average operators for vector valued functions

$$\begin{aligned} \{\{\mathbf{v}\}\} &= \frac{1}{2}(\mathbf{v}^+ + \mathbf{v}^-) \text{ in } \Gamma_I, & \{\{\mathbf{v}\}\} &= \mathbf{v}^+ \text{ in } \Gamma, \\ \llbracket \mathbf{v} \rrbracket &= (\mathbf{v}^+ \cdot \mathbf{n}^+ + \mathbf{v}^- \cdot \mathbf{n}^-) \text{ in } \Gamma_I, & \llbracket \mathbf{v} \rrbracket &= \mathbf{v}^+ \cdot \mathbf{n}^+ \text{ in } \Gamma, \end{aligned}$$



for vector valued functions  $\mathbf{v}_h \in \mathbf{V}_h^p$ .

Following Hartmann and Leicht,<sup>15,16</sup> we arrive at the semi-discretization of Equation 4 in element based form, find  $\frac{\partial \mathbf{y}_h}{\partial t} \in \mathbf{V}_h^p$  such that

$$\int_{\Omega} \frac{\partial \mathbf{y}_h}{\partial t} \cdot \mathbf{v}_h \, d\mathbf{x} - \int_{\Omega} (\mathcal{F}^c(\mathbf{y}_h) - \mathcal{F}^v(\nabla \mathbf{y}_h)) : \nabla \mathbf{v}_h \, d\mathbf{x} + \sum_{\kappa \in \mathcal{T}_h} \int_{\Gamma} (\hat{\mathbf{h}} - \hat{\sigma}_h \mathbf{n}) \cdot \mathbf{v}_h \, ds + \sum_{\kappa \in \mathcal{T}_h} \int_{\Gamma} (\hat{\mathbf{y}}_h - \mathbf{y}_h) \otimes \mathbf{n} : \nabla \mathbf{v}_h \, ds = \mathbf{S}(\mathbf{y}_h), \quad (10)$$

$\forall \mathbf{v}_h \in \mathbf{V}_h^p$  and  $\hat{\mathbf{h}}$  is a consistent and conservative numerical flux associated with the convective operator. The numerical fluxes associated with the viscous operator,  $\hat{\mathbf{y}}_h$  and  $\hat{\sigma}_h$ , are defined following<sup>16</sup> as

$$\hat{\mathbf{y}}_h = \{\{\mathbf{y}_h\}\}, \quad \hat{\sigma}_h = (\{\{\mathcal{F}_{\nabla \mathbf{y}}^v \nabla \mathbf{y}_h\}\} - \eta \delta(\mathbf{y}_h)) \text{ in } \Gamma, \quad (11)$$

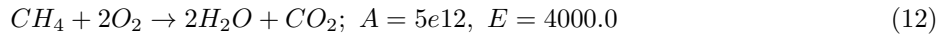
where  $\eta$  is a constant chosen such that stability is guaranteed.

## C. Temporal integration and simple chemistry model

The semi-discrete equations are discretized in time using a Standard-Strong-Stability-Preserving-Runge-Kutta method.<sup>17</sup> The temporal integration of the non-stiff convection operators is separated from the temporal integration stiff source term via Strang operator splitting. The temporal integration of the chemical source term is done implicitly via finite elements. Second order accurate integration is used for the explicit integration and up to fifth order accurate integration is used for the chemical source term.

### 1. Simple Chemistry Model

For this study we were interested in the flow characteristics of the flameless combustor, however, we introduced a simple single step reaction to simulate the heat release and concentration fluctuations and their interactions with the fluid mechanics. The single step reaction and Arrhenius rate parameters are shown below



where activation energy is  $E$ , in cal/mole, and  $A$  is the pre-exponential factor. The tailored modeled assured complete combustion of methane at various equivalence ratios for temperatures matching the operating experimental conditions ranging from 1000-1200 K within a residence time of 300 milliseconds. The residence time was estimated from preliminary cold flow simulations.

## D. Simulation Configuration

The hot bath, internal mixing bowl, fuel injectors, combustion region, and exhaust outflow were all accounted for in the considered simulation domain. Figure 2 depicts this simulation domain. The hot bath's bottom boundary is a subsonic stagnation boundary condition that is set to 1000 K, standard air concentrations, and 1.2 atm. All walls are treated with a slip wall boundary condition. The exhaust is modeled with a subsonic outflow condition with fixed pressure as the inflow characteristic. Not pictured is the injector boundary conditions which were also subsonic stagnation boundaries, with only methane at 1.1 atm and 300 K.

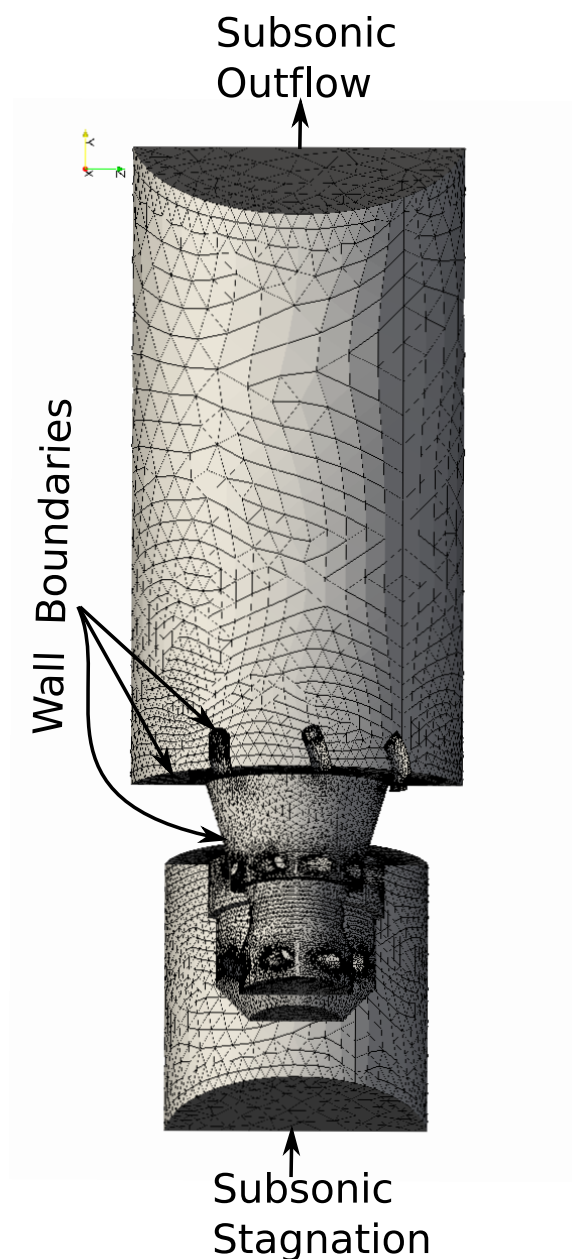


Figure 2. Simulation domain with labeled boundaries.

#### IV. Results and Ongoing Work

Here we present initial results using the simple methane model and the inviscid ( $D_k = \mu = \lambda = 0$ ) formulations of Equations 1-4. From preliminary solutions, we noticed that convective velocities ranged from 50-250 m/s within the simulation domain, which allowed for the removal of the diffusion terms to explore the convective structures within the combustor. Artificial viscosity was used to smooth discontinuities and regions of high shear using the shock capturing explained by Hartmann.<sup>18</sup> We first ran simulations until the maximum  $\text{CO}_2$  average within the combustion region of the domain stabilized to be near a mass fraction of 0.013. From that point, we considered the simulation to be fully developed and computed the various diagnostics presented within this manuscript. Here we present time-averaged data as well some dynamic data.

## A. Time Averaged Data

After the simulation reached a fully developed state, we ran the simulation for a window of physical time of 0.02 seconds. In that time we dumped data at a frequency of 25000 hz for a total of 500 three dimensional DG data dumps. Those 500 domains were averaged together via a running average. The flow characteristics of various regions of the experimental configuration within this section were computed directly from the running average data.

### 1. Fuel Injector

A parametric plane cut pictured in Figure 3(a) was used to sample the mid section of the fuel injectors. The bottom side of the plane runs radially outwards from the center of the the cylindrical combustor. The left side of the plane is at the same angle that the center point of the injector exit face is with respect of the flat surface around the base of the injectors,  $82^\circ$ . Throughout this paper, the bottom side direction is labeled as  $r$  as it is directly in line with the radial direction of the cylindrical combustor. The left side direction is labeled as  $t$  as it is a parametric linear coordinate. Figure 3(b) shows all six injector planes with average methane mass fraction,  $Y_{CH_4}$  displayed.

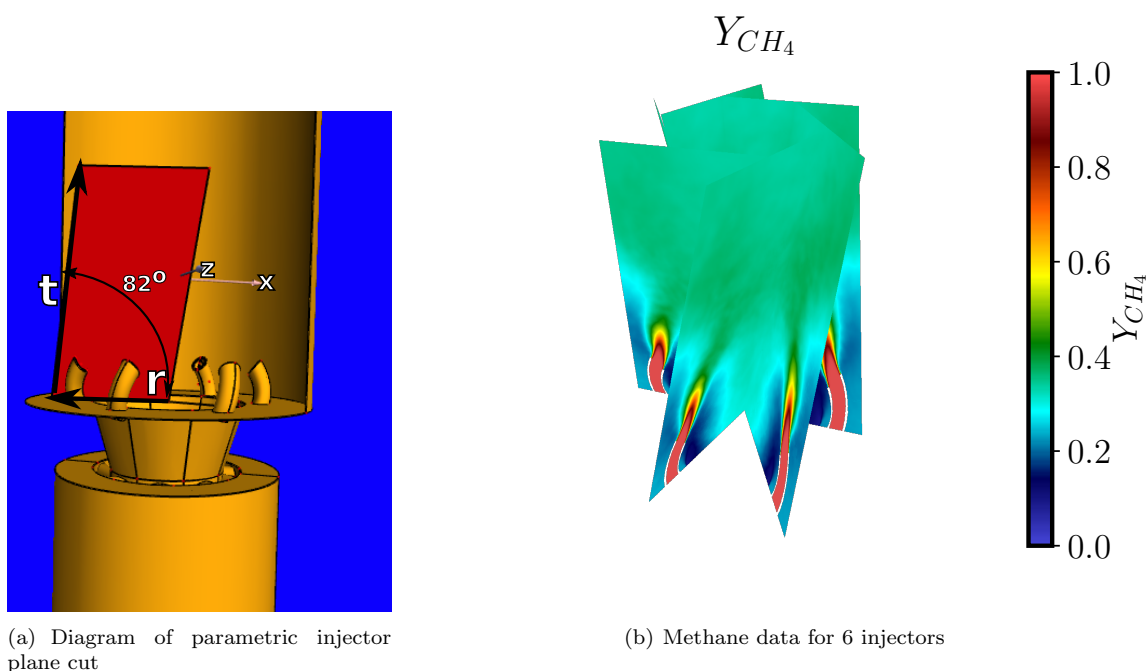
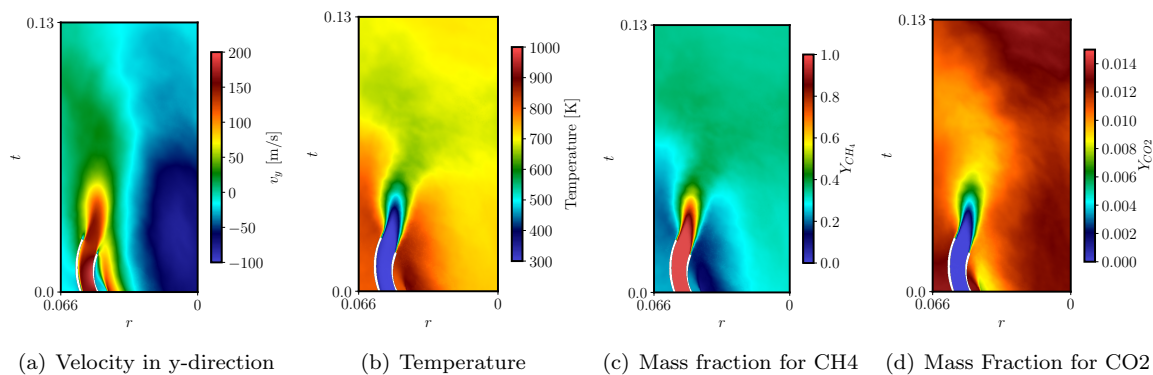


Figure 3. Details of the fuel injector plane cuts

Here we present the results from one of these six plane cuts as there was no meaningful difference between the averaged data of the six injectors. Figures 4(a), 4(b), 4(c), and 4(d) show the average value for y-direction velocity, Temperature,  $CH_4$  mass fraction, and  $CO_2$  mass fraction. The y-direction velocity profile shows that velocity ranges from -100 to 200 m/s in this region of the combustor. The y-direction velocity is positive nearest to the wall and at the injector exit. The areas closer to the center of the combustor center of negative direction velocity, which is an indicator of flow recirculation.



**Figure 4. Details of the fuel injector plane cuts**

Temperature, Figure 4(b), is highest near the swirler wall in the area near the base of the outside injector walls. Here the flow from the hot bath is transported from the swirler and transports the hot air to the cold methane injectors. The hot air mixes in the area near the exit of the injector causing a cooler temperature of approximately 600 K. Further downstream in the combustor the temperature begins to rise to about 750 K. Additionally the areas radially outward of the injector is a host for high temperatures of about 900 K, due to hot air convected around the injectors from the swirler region.

The averaged fuel injection of methane, Figure 4(c), penetrates radially inward. The mass fraction of methane reduces to 0.50 in within two Diameters normal of the injector exit with injection with higher injection existing more radially outward. The methane is spread throughout the entire region above the injection and is spread into the center of the combustor where it is convected into the the swirler region. Methane is lowest near the base of the outside injector walls where the air is convected from the swirler.

CO<sub>2</sub>, Figure 4(d), is found well mixed through the entire combustor. The highest amounts are found radially outside the injectors and within the center of the combustor. Additionally CO<sub>2</sub> has high concentrations along the swirler wall leading up the injector to where methane is injected. This indicates that combustion products are recirculated to meet the methane injection.

## 2. Penetration Plane

Figure 5(a) shows a plane located at the highest point of the fuel injectors. The plane is normal to the y-direction, existing in the x and z directions of the combustor. Throughout this manuscript we will use the metric D to represent the injector diameter of 8 mm. Data was sampled along this plane and 9 others that are 1.25 D apart in the y-direction of the combustor. These data planes are illustrated in Figure 5(b) with the injector planes from the previous section. The averaged data sampled in these planes highlight various features of the combustor, such as methane penetration distance, species spread, swirl, and vertical recirculation.

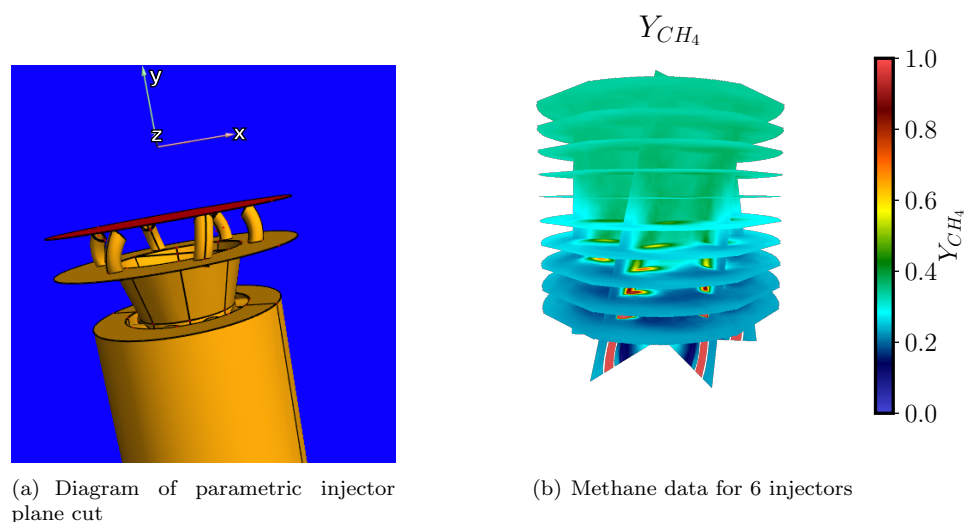


Figure 5. Details of the fuel injector plane cuts

Figure 6 shows the average methane concentration at six penetration planes starting from the lip exit, 6(a), and ending at 6.25 D, 6(f). Each methane injector is rotationally symmetric for the averaged field. The peak methane concentration for each of these planes as a function of radial distance is shown in Figure 6(g). As distance from the injector is increased the methane spreads towards the center of the combustor. At distance greater than 6.25 D the methane is well spread to a mass fraction value of 0.3 and is not reported within the figures.

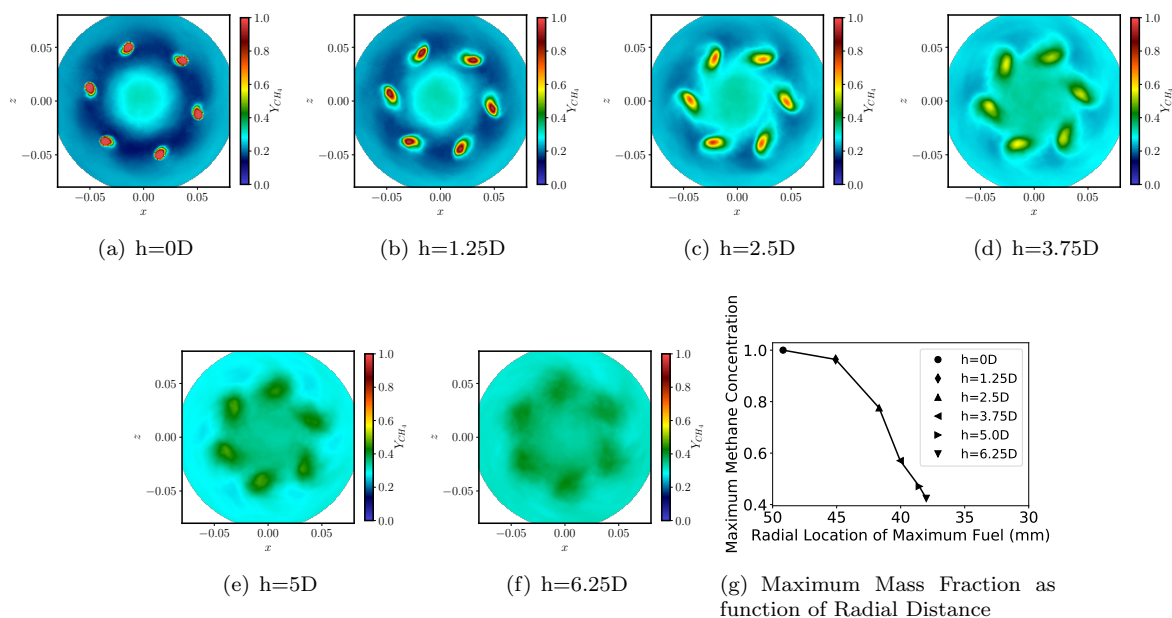
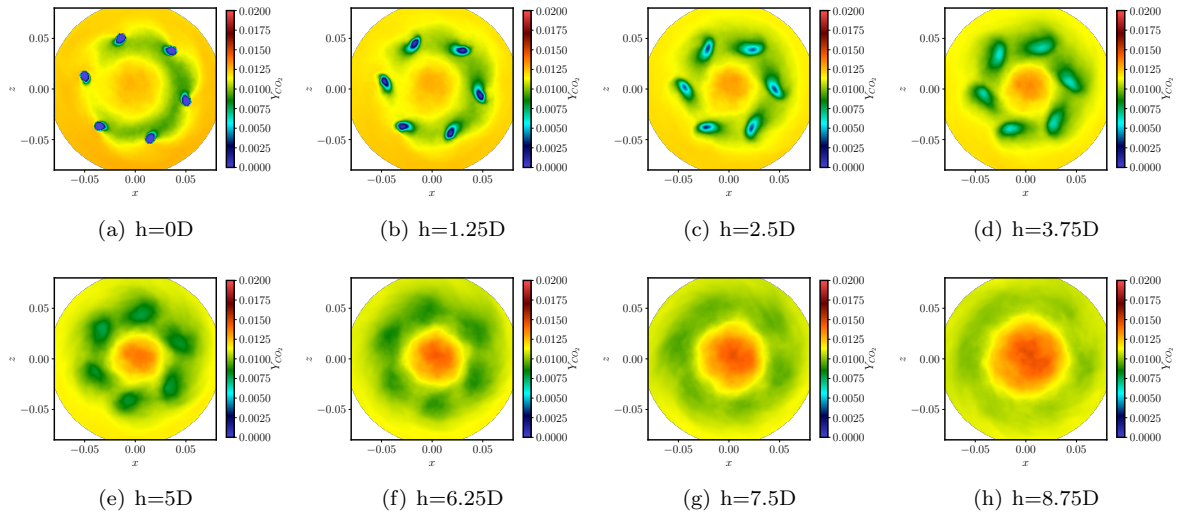


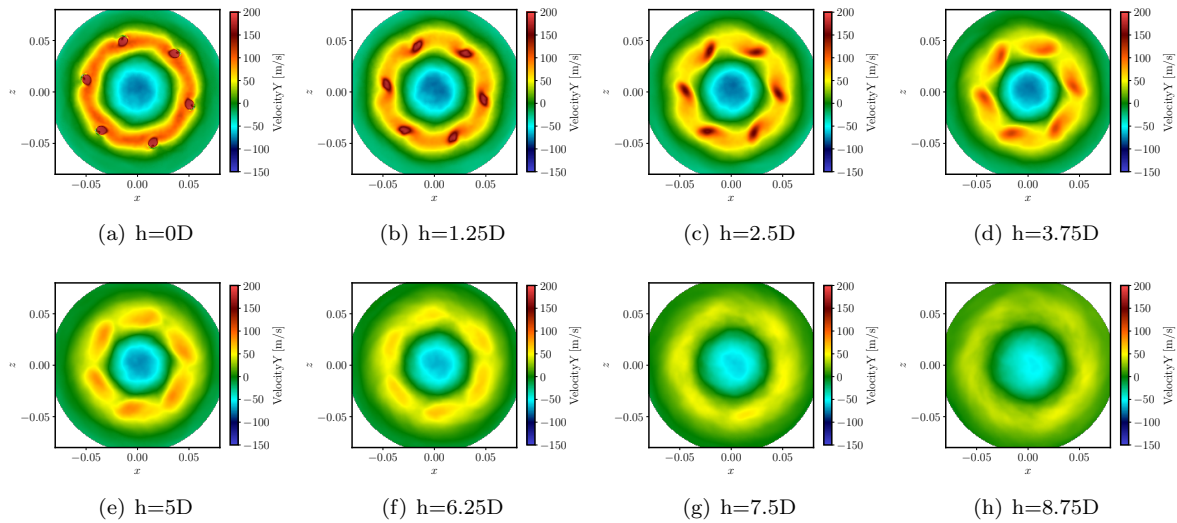
Figure 6.  $CH_4$  Mass fraction at different plane cuts increasing in distance,  $h$ , away from injectors. Distance normalized by injector diameter,  $D=8$  mm

The  $CO_2$  mass fraction is shown in Figures 7(a)-7(h).  $CO_2$  is largest both outside and inside the regions less than 2.5D from the injector exit planes. The center of the combustor hosts the largest amount of  $CO_2$  in distances greater than 2.5 D from the injector exit plane. The planes farther downstream of the injector host a spreading of  $CO_2$  in a uniform radial direction.



**Figure 7.** CO<sub>2</sub> mass fraction at different plane cuts increasing in distance,  $h$ , away from injectors. Distance normalized by injector diameter,  $D=8$  mm

The velocity in the  $y$ -direction of the combustor is shown in Figures 8(a)-8(h) for the penetration plane cuts. For planes near the injectors, the average velocity changes concentrically in several areas. The innermost region hosts a negative velocity peaking at  $-60$  m/s, whereas the regions near the injectors host a positive velocity of  $200$  m/s. For the planes in areas farther from the injector the positive velocity from the injectors is diminished to less than  $50$  m/s while the central negative velocity remains at the relatively the same magnitude of the lower previous planes.



**Figure 8.** Velocity in  $y$ -direction at different plane cuts increasing in distance,  $h$ , away from injectors. Distance normalized by injector diameter,  $D=8$  mm

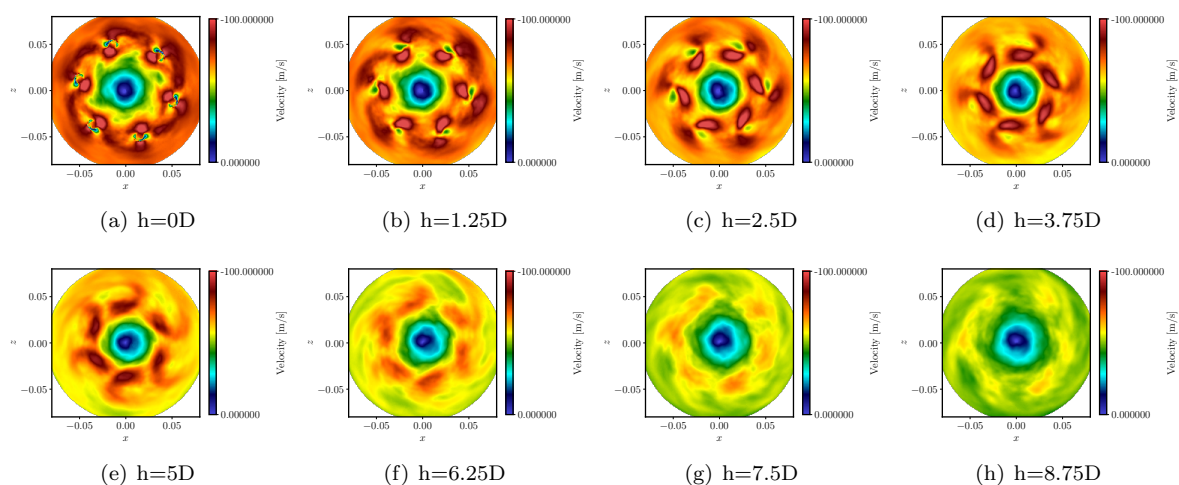
The combustor's swirl velocity was calculated by

$$v_{\theta} = v_x \sin(\theta) + v_z \cos(\theta), \quad (13)$$

where  $\theta = \arctan \frac{z}{x}$  and is shown for the penetration plane cuts in Figure 9(a)-9(h). Here, a positive swirl is in the clockwise direction. There is a consistent negative swirl for all average plane cuts that is stronger for the planes closer to the exit of the injectors. For planes closer than  $5D$ , the swirl is minimum in the center and just outside of the fuel injector plumes. The swirl is maximum in locations above the fuel plumes.



Planes farther than 5D do not have minimums in the locations that are present in lower planes that are from the injector influence.

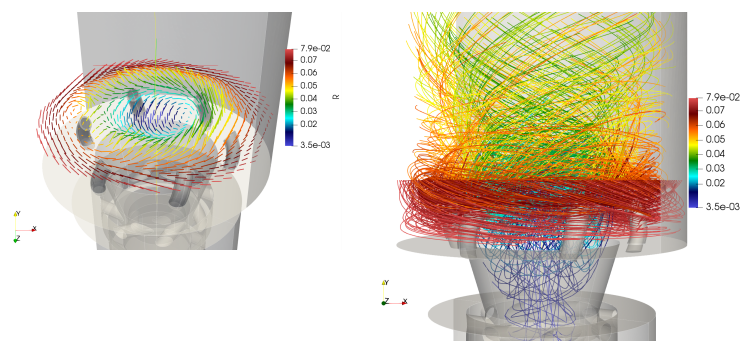


**Figure 9.** Swirl Velocity,  $v_\theta$ , at different plane cuts increasing in distance,  $h$ , away from injectors. Positive swirl is in the clockwise direction. Distance normalized by injector diameter,  $D=8$  mm

## V. Conclusion and Future Work

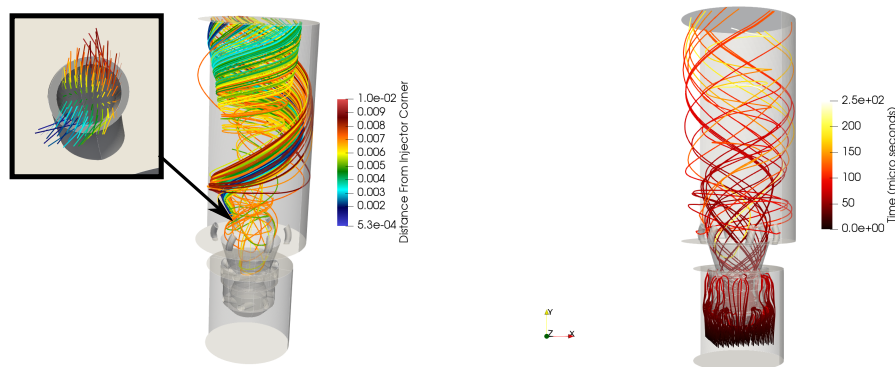
### 1. Average Flow Streamlines

We used the averaged three dimensional data as a steady state solution and seeded lagrangian tracer particles at various locations to better visualize the three dimensional flow physics. The propagation of these seeded particles are blind to diffusion, both physical and artificial, and thereby are effectively the averaged state streamlines. Figure 10 shows the streamlines generated from 250 particles seeded 1.25 D above the fuel injectors in a plane normal to the y-direction. The streamlines are colored by the radial location of the primary seeded location, which is shown in the left image contained within Figure 10. The image to the right is the propagation of the particles after 300 milliseconds. The outer most particles stay within the combustor by circulating around the injectors. The mid region particles (green-yellow) move vertically away in a helical motion. The inner most particles move downwards into the swirler and move in a recirculating motion.



**Figure 10.** Particle Pathlines starting from a plane just above the injectors colored by distance from center of the cylindrical combustor

Figure 11(a) shows streamlines generated from particles seeded from the injector's exit face. As shown in the highlighting box, the streamlines are colored by exit plane seeded position. The inner most (closest to swirler wall) seeded particles exit the combustor in a helical pattern. The outer most (farthest from the swirler wall) also exit the combustor in a circular pattern. The particles ranging from the middle to outer-middle penetrate into the recirculation zone and travel into the swirler.



(a) Particle path lines colored by point of origins distance from the center of the injector exit face. (b) Particle path lines colored by time.

**Figure 11. Particle Pathlines starting from injector, 11(a), and hot path, 11(b).**

Figure 11(b) shows streamlines for particles seeded within the hot air bath. The streamlines are colored by time. The streamlines feed into the swirler through the swirler veins, travel along the swirler walls, and up into the combustion region where they exit the domain in a helical pattern. Some of the streamlines recirculate within the combustion region as shown by having streamlines of time 250 milliseconds within the swirler, where as most streamlines have exited the combustor by then. Some streamlines also move radially outward of the injectors.

## A. Dynamic Flow Physics

This section presents the details of data at specific times sampled along the same planes previously presented. We cover the qualitative assessments of the complex, unsteady, flow features that are present within the combustor. At the conference we will continue the analysis of the time dependent data as well as present movies that show more detail that can not be contained within this manuscript.

### 1. Injector

Figure 12 presents  $\text{CH}_4$  mass fractions with the same geometric plane definition that was explained in Section A-1. We have data sampled at 50  $\mu\text{s}$  increments for this plane cut. For this manuscript, we chose a point in time where the planar cut had a comparatively elongated plume of methane injection, Figure 12(a), and report five more plane cut samples at 200  $\mu\text{s}$  intervals of increasing time, Figures 12(b)-12(h).

There are eddies on scales smaller than the averaged flow characteristics revealed in the previous section. The functionality of these scales and how they factor into the behavior of the flameless combustor is an ongoing objective of our research, but these small scales are indicative of unsteady convective transport that requires this detailed analysis understand how the recirculation of methane combustion is established within this combustor.

At both 200 and 1000  $\mu\text{s}$  a methane pocket is visibly removed from the methane plume at approximately 2.5 D from the exit of the injector. After the pocket is removed, the plume grows back out to a longer length. We are not sure if this pocket removal is cyclic in nature or if the results presented here are merely fortuitous, however, we would like to note that the repetition of the is pocket removal does appear to happen in a cyclical mode in the long time window we have analyzed. For this reason, we will perform a fast fourier transform of the time-series data and present the appearance of any modes at the 2018 ASM-AIAA conference.



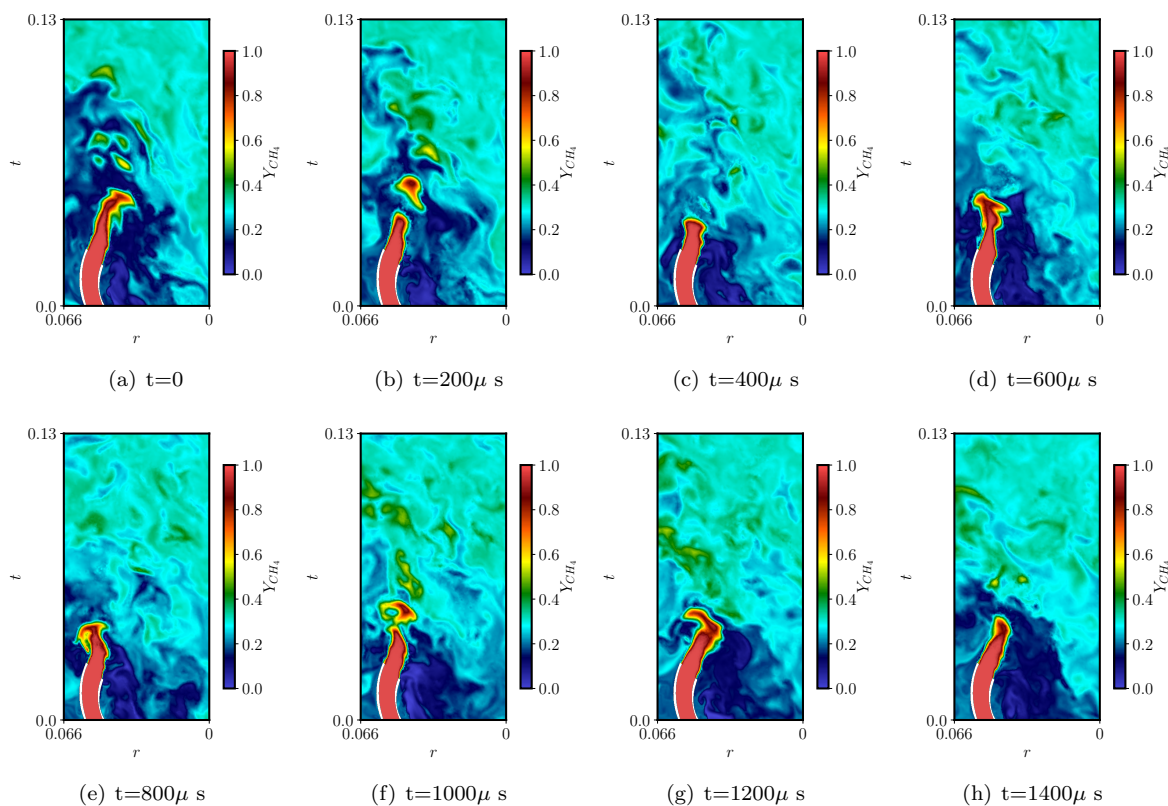


Figure 12. Mass fraction of  $\text{CH}_4$  at several times sampled through the fuel injector plane.

## 2. Penetration Planes

The penetration planes are shown at an initial time,  $t=0$ , where we found methane was at its peak in planes up to 5.0 D, which should be similar to the injector plume at one of its maximum plume heights, like that at  $t=0$  in Figure 12. We then display the plane cut at increasing 200  $\mu\text{s}$  intervals for distances of 0, 1.25D, 2.5D, 3.75D, and 5D away from the injector exit. There are small scale, unsteady structures in all of these plane cuts that are indicative of mixing methane towards the center recirculation zone. This is made evident by a swirling pool of fuel in the center of the planes that is fed by the injectors in planes 0D and 1.25D. The pool has a connection to several injectors, and the connected injectors change with respect to time. This indicates that the unsteady structures are convecting methane radially inwards into the recirculation zone and that there may be some possible cyclic modes for the injectors that contribute fuel the most.

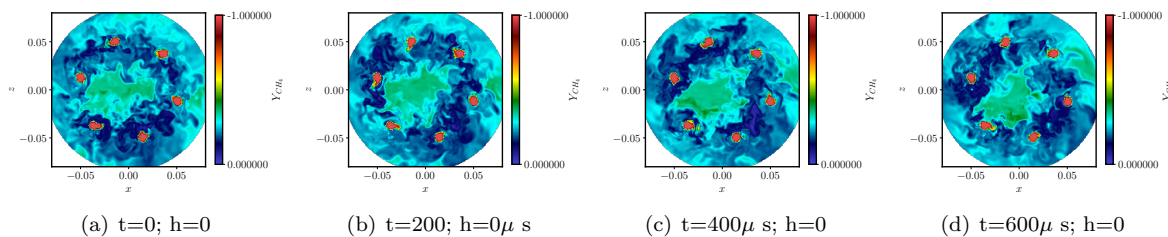
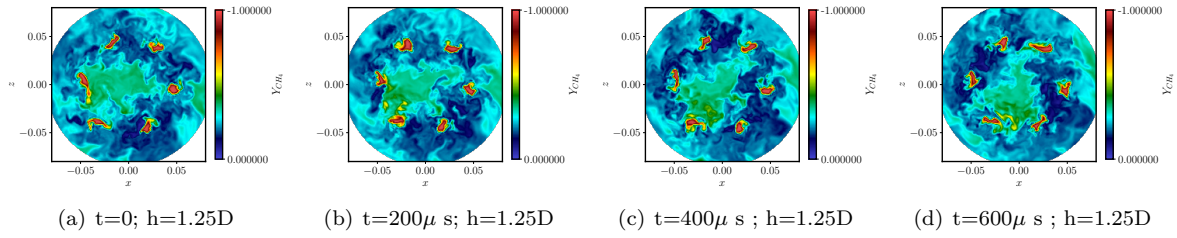
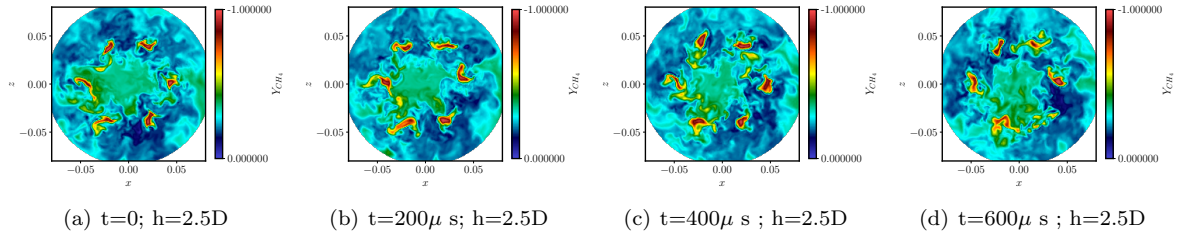


Figure 13. Instantaneous methane mass fractions at height of  $h=2.5$  D, distance normalized by injector diameter,  $D=8$  mm

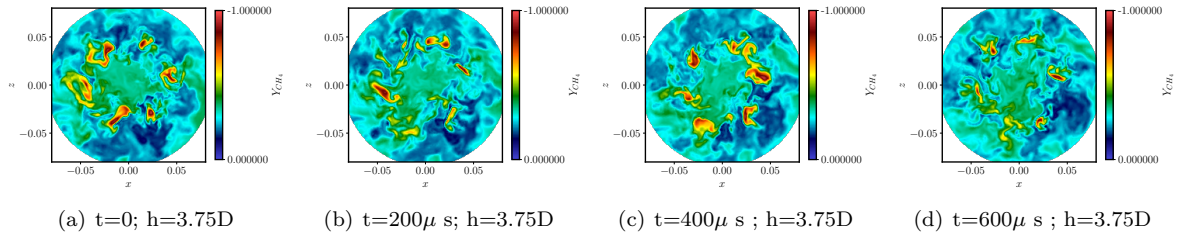


**Figure 14.** Instantaneous methane mass fractions at height of  $h=2.5$  D, distance normalized by injector diameter,  $D=8$  mm

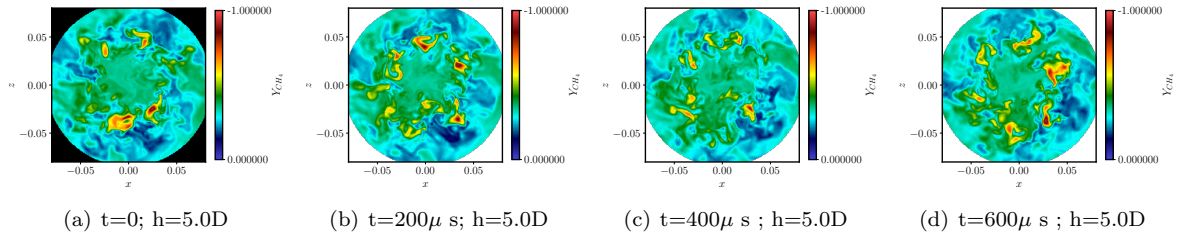
At plane cuts higher than  $2.5$  D, the six peak spots from the fuel injection are not always present. For instance, Figure 15(a) has six peaks but then Figure 15(b) has three peak methane values that exceed a mass fraction of  $0.8$ . The six peaks then return in Figure 15(c). This is indicative of the fuel pocket ripping off that was scene in the injector plane in Figure 12. By the time of the presentation we will perform a time series analysis to see if any modes match these peaks disappearing and reappearing. These modes will be compared to modes found in the injector plane data.



**Figure 15.** Instantaneous methane mass fractions at height of  $h=2.5$  D, distance normalized by injector diameter,  $D=8$  mm



**Figure 16.** Instantaneous methane mass fractions at height of  $h=3.75$  D, distance normalized by injector diameter,  $D=8$  mm



**Figure 17.** Instantaneous methane mass fractions at height of  $h=5.0$  D, distance normalized by injector diameter,  $D=8$  mm

## VI. Conclusion and Future Work

A three dimensional discontinuous-Galerkin finite element code for reacting fluid dynamics, Propel, was used to simulate the University of Cincinnati's experimental combustion configuration that was shown in previous works to host flameless combustion. A simple one-step reaction mechanism was used to simulate combustion product development and convection as well as heat release. Simulations were run until a stable, developed reacting solution was obtained. We then ran simulations for a physical time frame of 0.02 seconds, where both averaged and time-dependent data were collected and presented within this manuscript.

The averaged data presented here revealed that there is a consistent recirculation zone within the center of the combustor. Cold methane fuel is fed into this recirculation zone and products formed from joining hot air are recirculated to the base of injectors, which is fundamental and necessary for flameless combustion processes. Averaged penetration planes revealed that injected methane moves radially inward and that  $\text{CO}_2$  is formed largest within the recirculation zone. Also, a persistent counter clockwise swirl was present that encouraged mixing but was obstructed by the fuel injection.

Averaged streamline data also revealed the important overall flow features of this combustor. Seeded particles were used to reveal that other dominant zones exist within the combustor such as a recirculating zone that is outside of the injectors, and a zone inbetween injectors and the center that transports fluid in a helical form out of the combustor. Seeded particles that started within the injector revealed that pure convection alone does allow a few streamlines to penetrate into the recirculation zone. Additionally, seeded particles in the hot bath reservoir revealed that the hot air is transported close to the walls of the swirler and follows the dominant helical structure that convects fluid out of the combustor.

Time-dependent data was also reported that showed fine scale structures in both injector and penetration planes. These fine scale structures had notable reoccurring phenomena, such as a methane pocket that is ripped off and is then followed by a growth of the methane plume. Additionally, the penetration planes revealed that these fine scale structures contribute to the mixing of methane into the center recirculating zone. Techniques such as a Fast-Fourier Transform of the planar data in time will be used to establish important cyclic modes and presented at the 2018 ASM-AIAA conference.

The simulation of this combustor revealed a variety of interesting structures and important zones that gives some insight into how flameless combustion was achieved by Gutmark et al<sup>6-11</sup> at the University of Cincinnati. This work revealed that convective structures alone can lead to the establishment of a recirculation zone. Future work will enable detailed species diffusion in the Propel simulations as well as include more detailed chemical models. The chemistry in flameless combustion is without a question a driving factor for a sustained flameless combustion system, so the proper prediction and simulating of radical species production within this combustion configuration must be considered before. Additionally, we will analyze the effects of geometric changes noticed by Gutmark et al. which will allow us to optimize this combustor configuration for more effective flameless combustion.